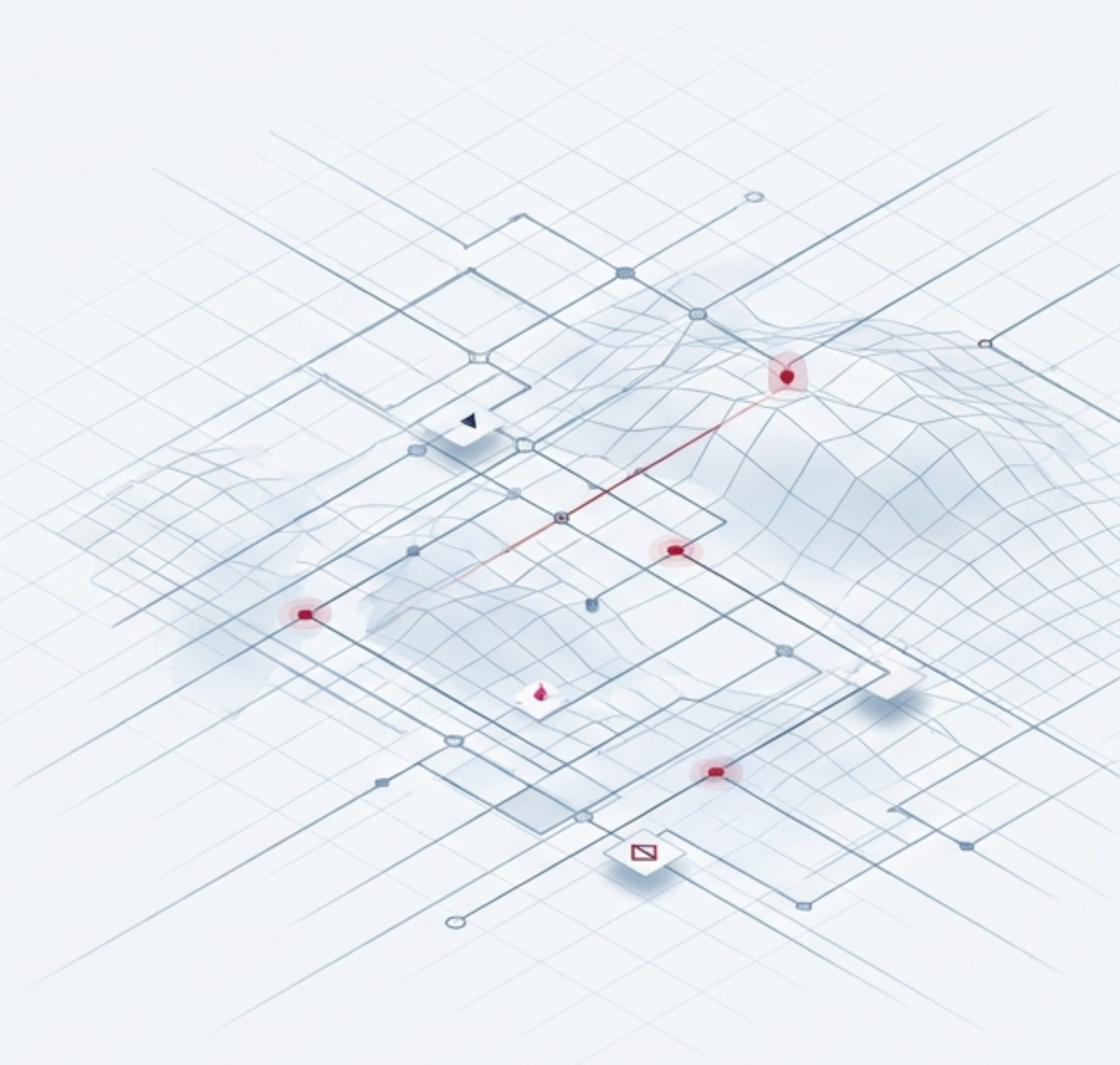


AI Powered Financial Fraud Detection & Risk Intelligence System

Machine Learning | Explainable AI |
Risk Intelligence | FinTech Analytics

Presented by Alin Kumar



Executive Summary



Business Problem

Rule-based detection systems are **failing** to catch evolving **cyber-threats**, resulting in millions lost daily to financial fraud and creating high operational friction from false alarms.



Project Goal

Develop an **Explainable AI (XAI)** risk intelligence system capable of dynamically identifying **fraudulent** patterns while strictly **minimizing false positives** and missed fraud.

99.56%
ROC-AUC

Key Achievements

The deployed system achieves an **88.52% F1 Score**, capturing hidden mathematical anomalies with unprecedented precision.

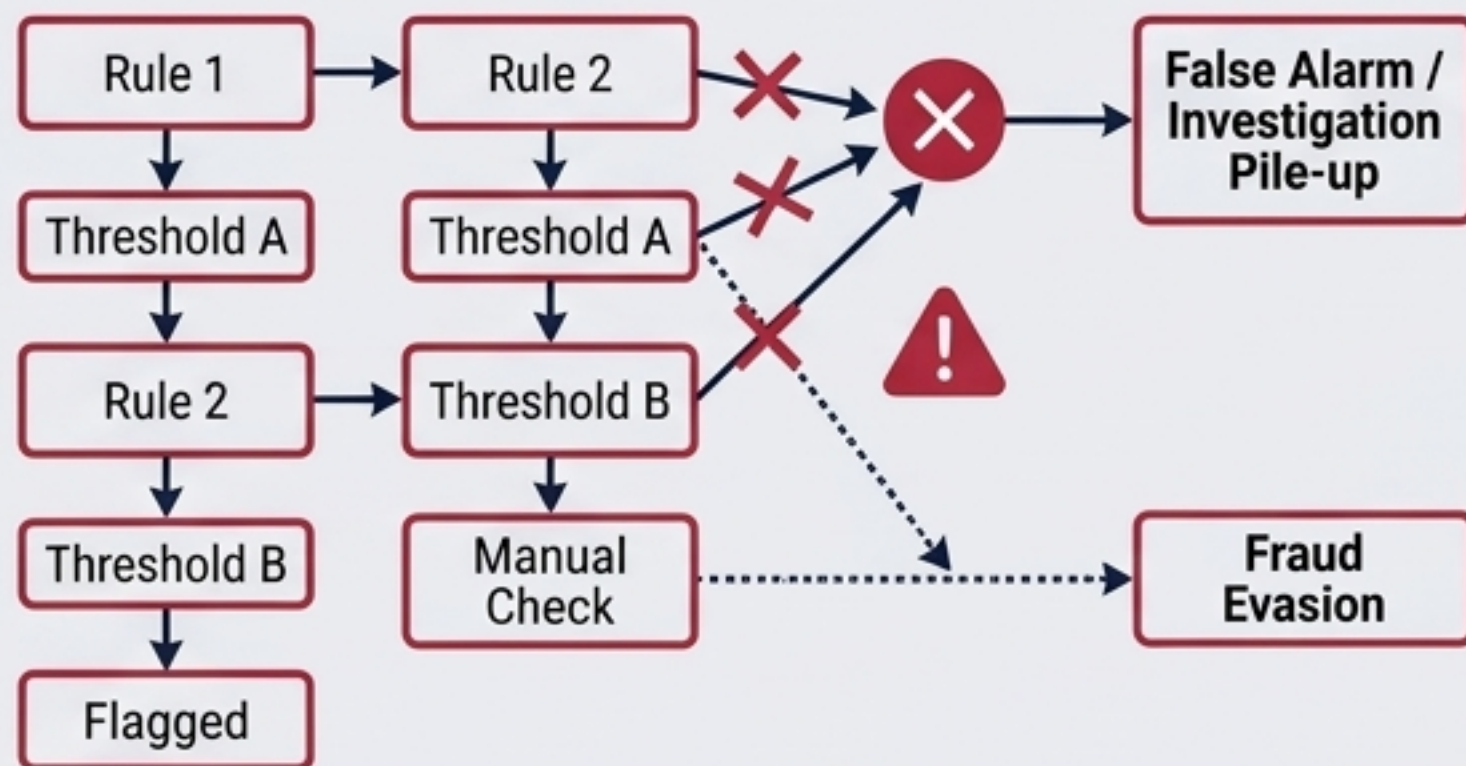


Expected Impact

Significant **reduction in financial risk** exposure, decreased manual workload for Fraud Investigation Teams, and frictionless experiences for **99.89%** of legitimate customers.

The Business Problem & Cost of Inaction

The Old Paradigm: Static Rules



High False Alarm Rates

- Rigid thresholds flag too many **genuine** transactions, causing friction and customer churn.

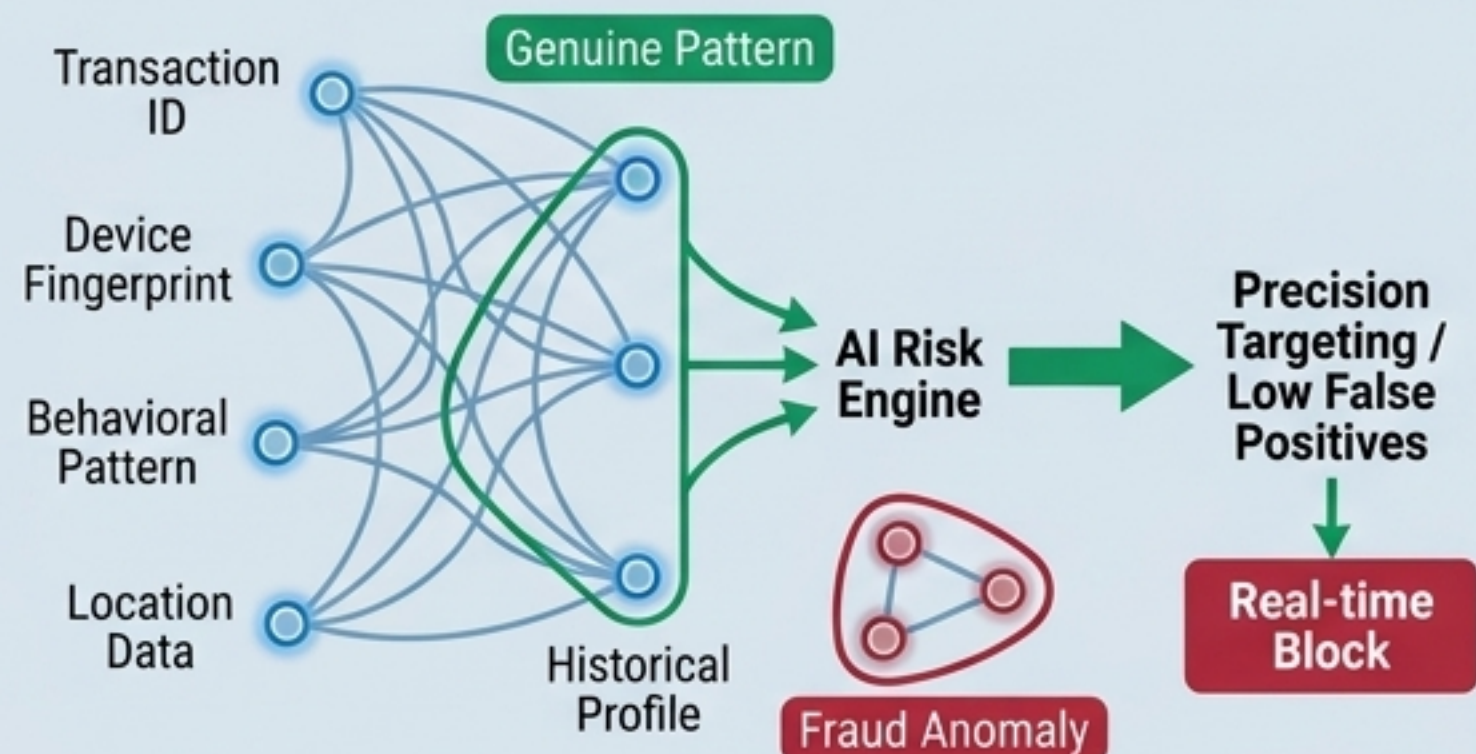
Evasion Tactics

- Fraudsters easily **reverse-engineer** and **bypass** static limits, constantly adapting their methods.

Operational Bottlenecks

- Investigation teams are **overwhelmed** by **noise** and manual reviews, increasing response times.

The New Paradigm: AI Risk Intelligence



Pattern Recognition

- Identifies complex, multi-variable fraud signatures by analyzing vast datasets and correlations.

Dynamic Adaptation

- Evolves continuously with new transaction data, **self-learning** to stay ahead of emerging threats.

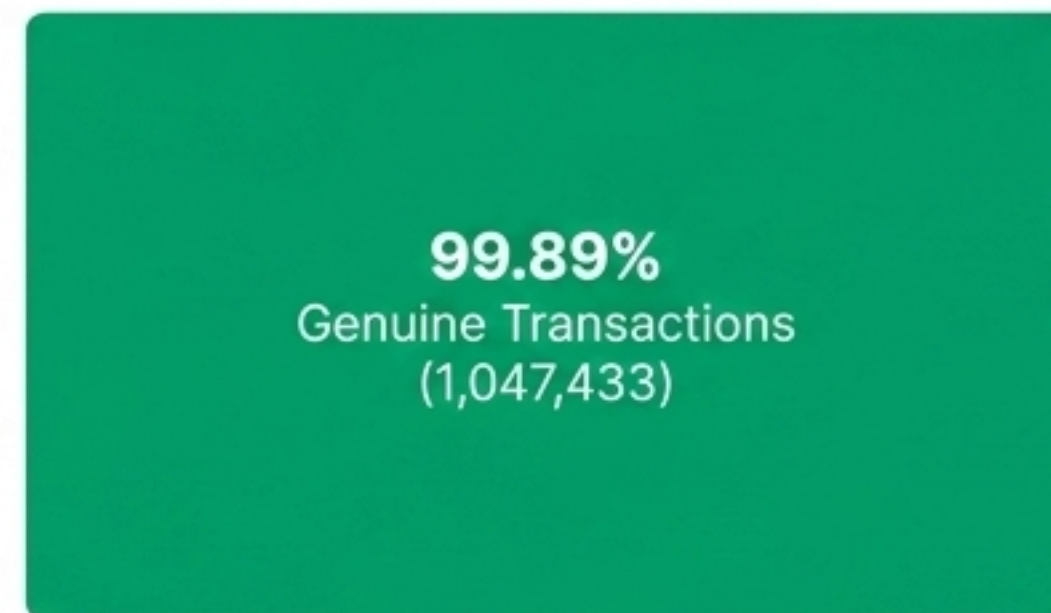
Precision Targeting

- Drastically **reduces false positives**, isolating true threats while enabling frictionless experiences for legitimate users.

Data Landscape & Intelligence Signals



The Imbalance Challenge



0.11%
Fraudulent Transactions
(1,142 cases)

Insight: The model must pinpoint highly isolated events without misclassifying the overwhelming majority of safe transactions.

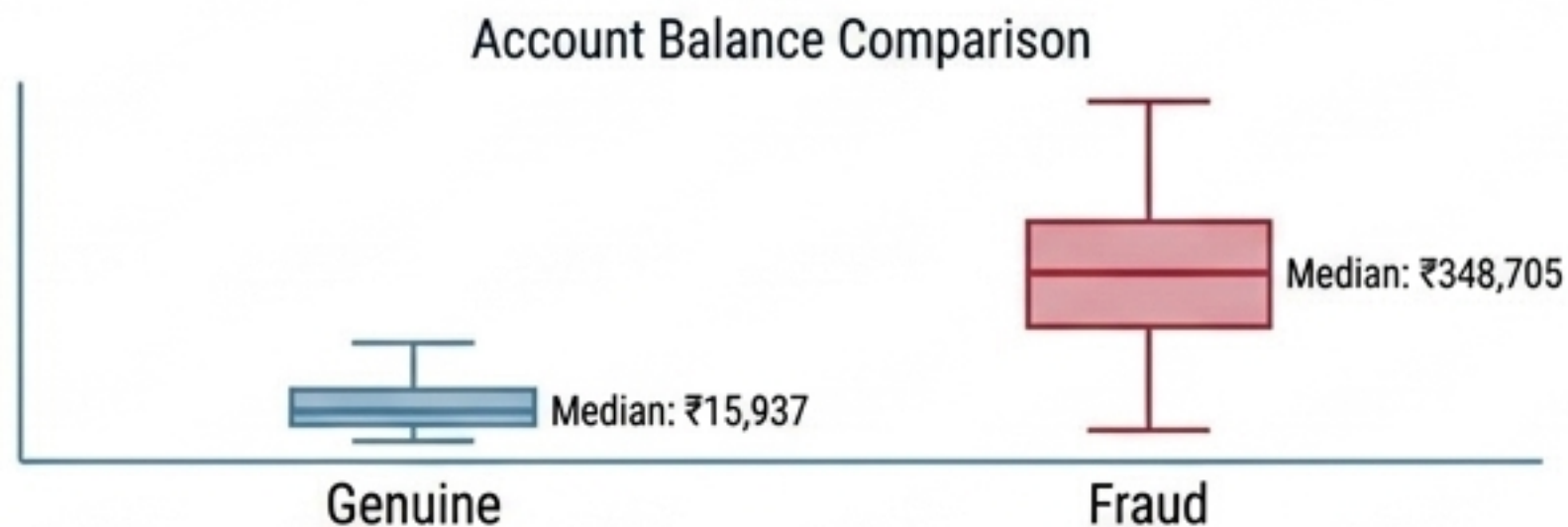
Exploratory Data Analysis: Signal Detection

Transaction Channel Risk



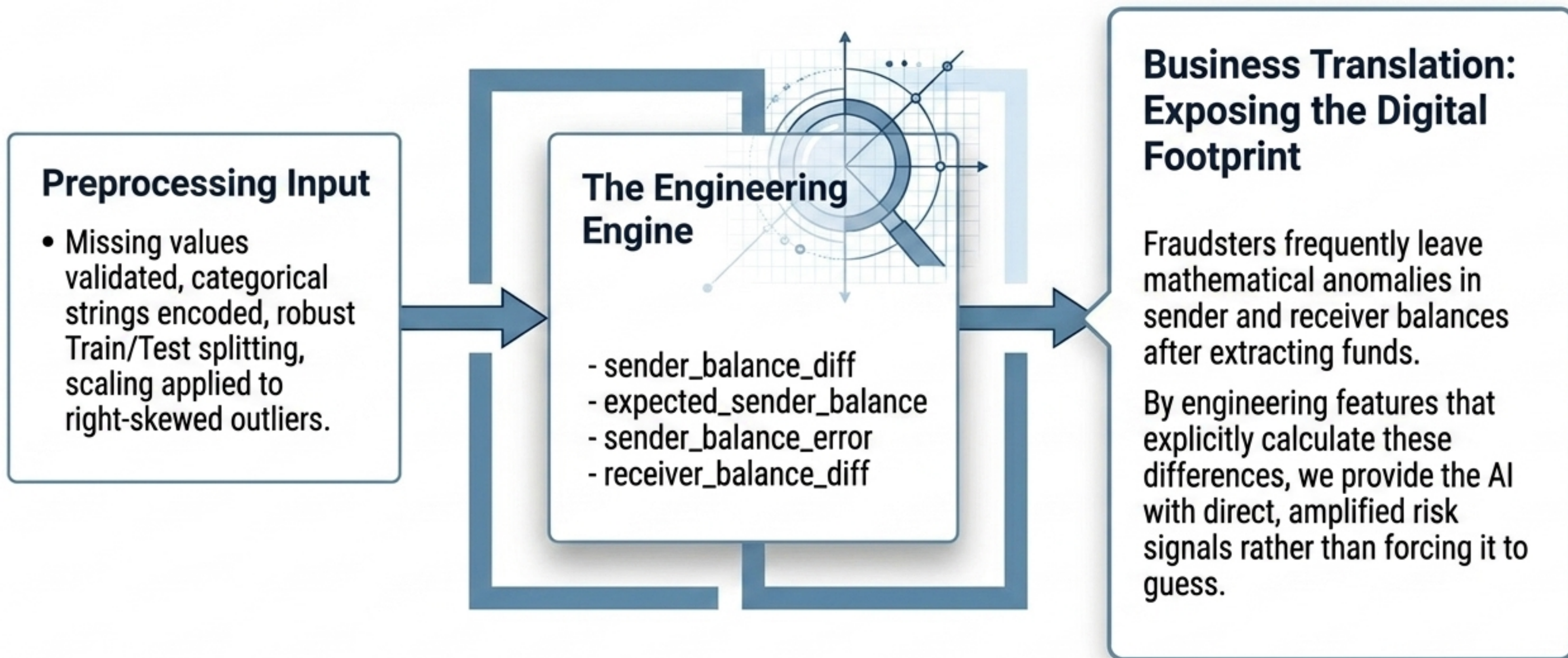
- **Observation:** 100% of detected fraud occurs exclusively in CASH_OUT and TRANSFER channels.
- **Insight:** Fraudsters extract liquidity via specific, rapid-exit channels.
- **Decision:** Heavily weight these transaction types as primary risk indicators.

The Balance Anomaly



- **Observation:** Fraudulent transactions originate from accounts with a median balance of ₹348,705, compared to just ₹15,937 for genuine.
- **Insight:** Threat actors actively target high-liquidity accounts for maximum extraction.
- **Decision:** Balance-related features require advanced scaling and explicit interaction mapping.

Feature Engineering & Preprocessing Pipeline

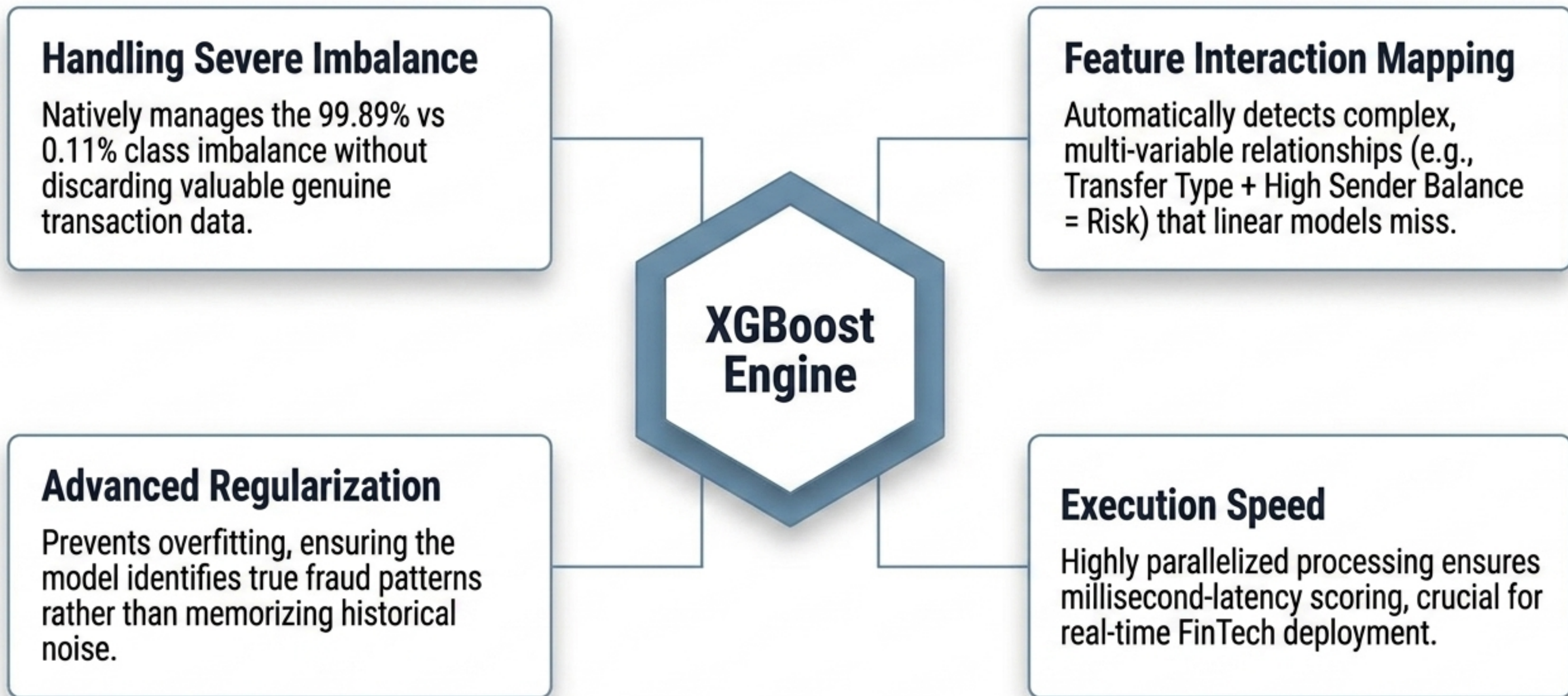


Baseline Machine Learning Models

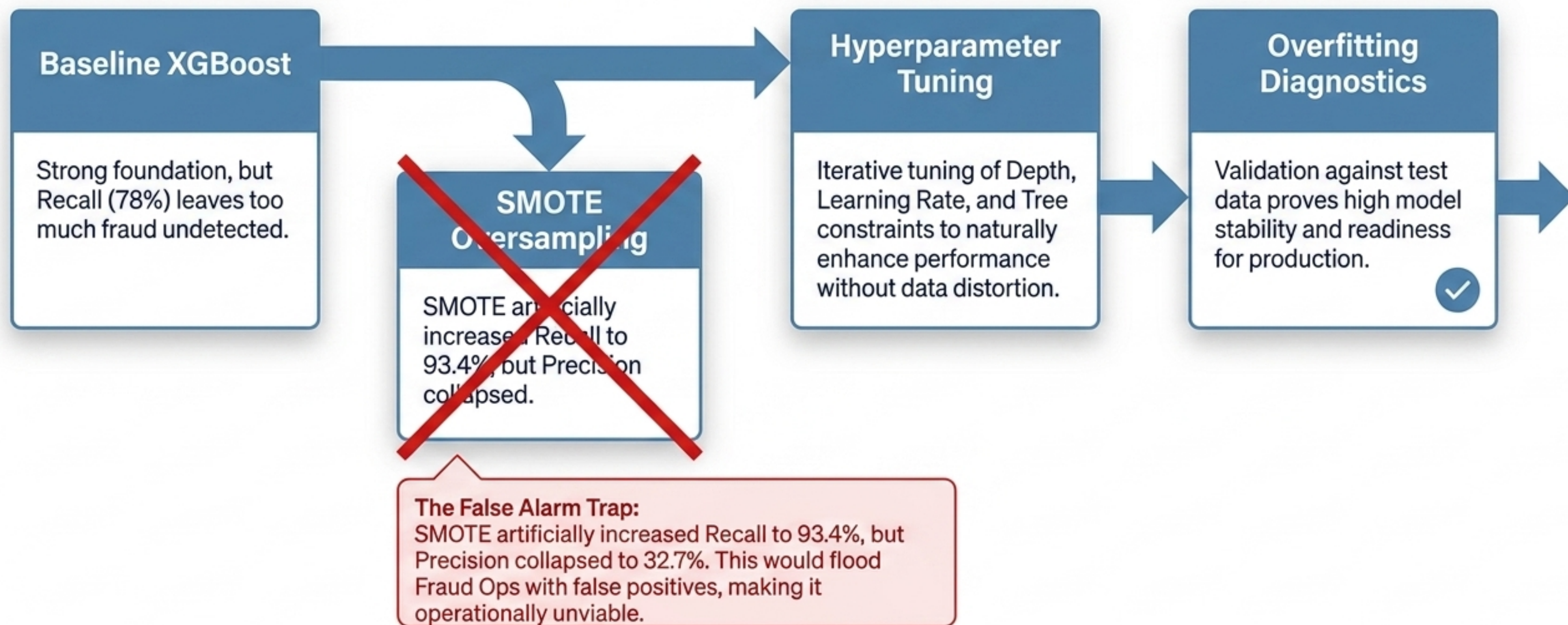
	Comparison Matrix			
Model	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	Fails to capture non-linear fraud patterns			
Decision Tree	80.8%	75.8%	78.2%	87.9%
Random Forest	98.8%	75.4%	85.5%	97.1%
Gradient Boosting	18.8%	31.5%	23.6%	55.1%
CHAMPION Baseline XGBoost	95.1%	78.0%	85.7%	98.3%

Conclusion: XGBoost establishes absolute dominance in F1 and ROC-AUC, providing the strongest mathematical foundation for further optimization.

The Champion Model: Why XGBoost Won

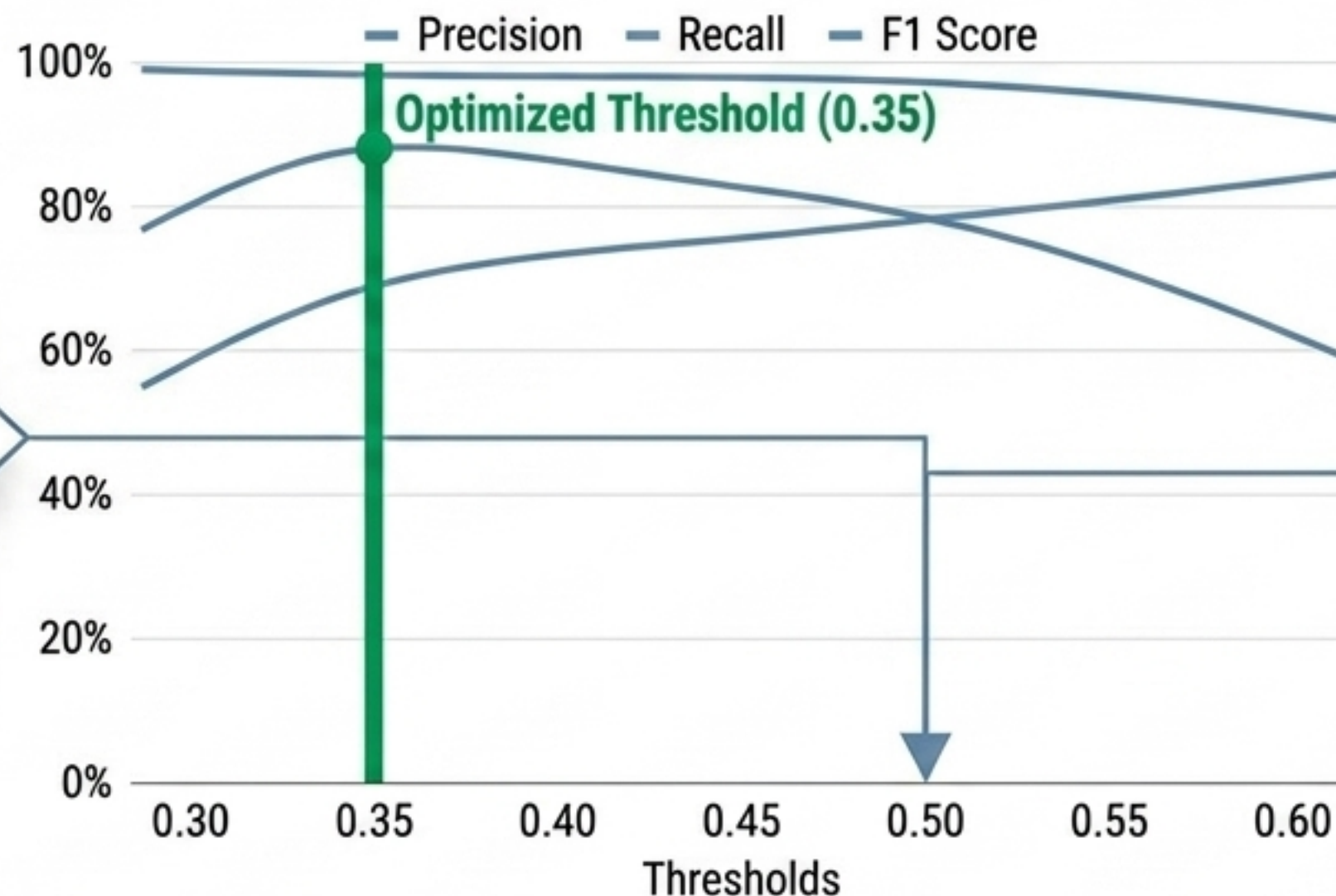


The Experiment Journey & Optimization Pipeline



The Goldilocks Zone: Threshold Optimization

"Business Friction vs. Risk" slider



Default Threshold = 0.50

- Recall stuck at 78.5%. Too conservative. Leaves significant fraud undetected to protect precision.

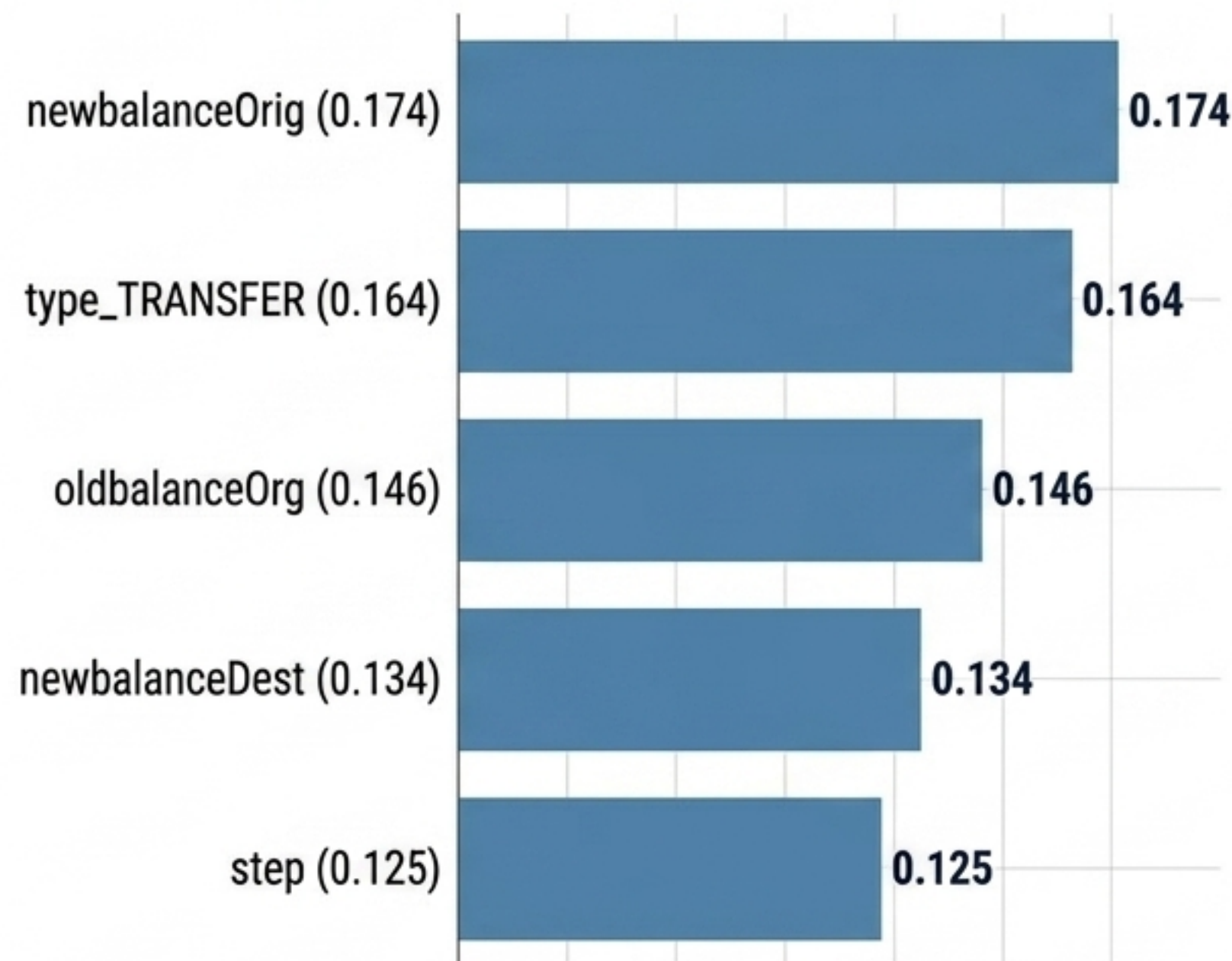
The Goldilocks Zone = 0.35

- Maximum F1 Score achieved (88.52%).
- Precision remains incredibly high (97.37%), while Recall jumps to 81.14%.

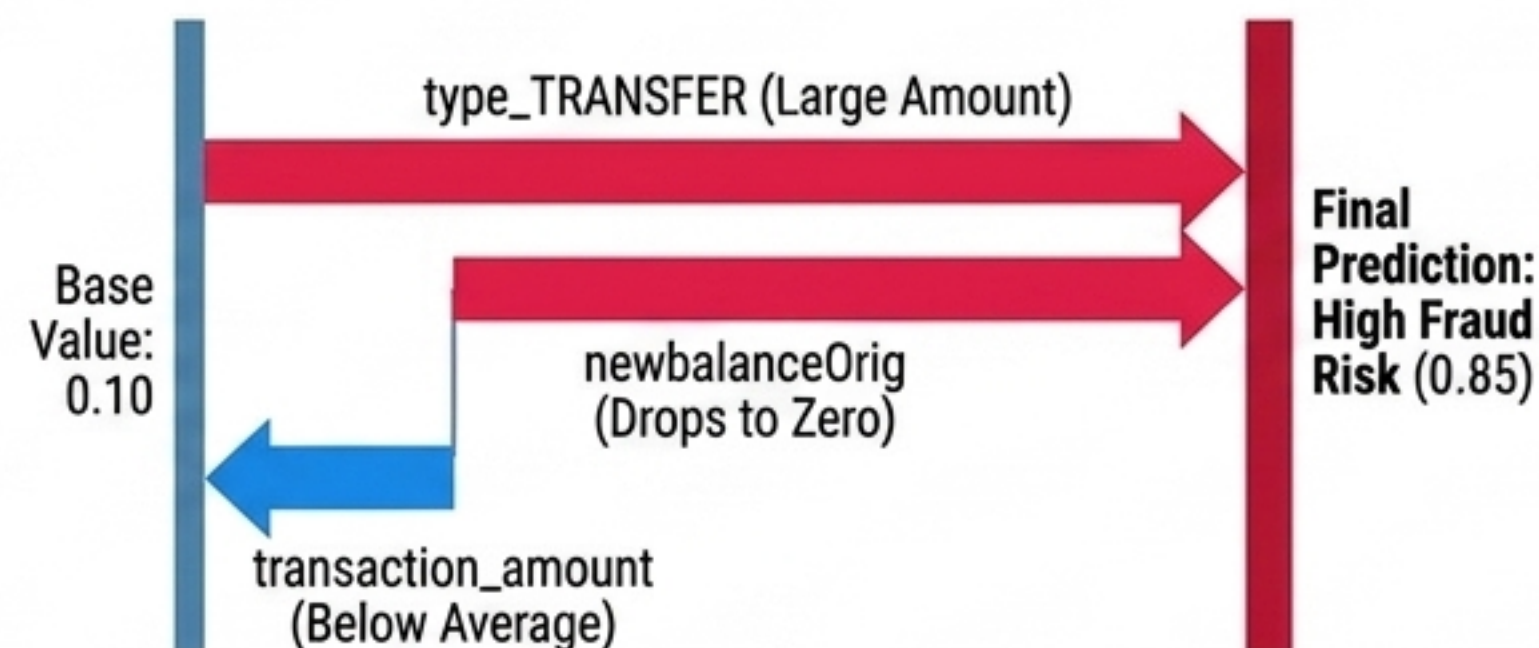
Executive Takeaway: Lowering the confidence threshold to 0.35 mathematically optimizes the system to catch more fraud without overwhelming investigators with false alarms.

Explainable AI (SHAP) - The Investigator's Lens

Global Feature Importance



The Investigator's Lens (SHAP)



What is SHAP?

Transforms AI predictions from a Black Box into Transparent Evidence.

Operational Use Case

When a transaction is flagged, Investigators instantly see exactly why (e.g., Alert triggered because Sender Balance dropped to zero during a large TRANSFER).

Value

Reduces manual investigation time from hours to minutes.

Final Deployed Model Performance (Optimized Threshold = 0.35)

ROC-AUC: 99.56%

Near-perfect class separation capability

F1 Score: 88.52%

The ultimate balance of precision and recall

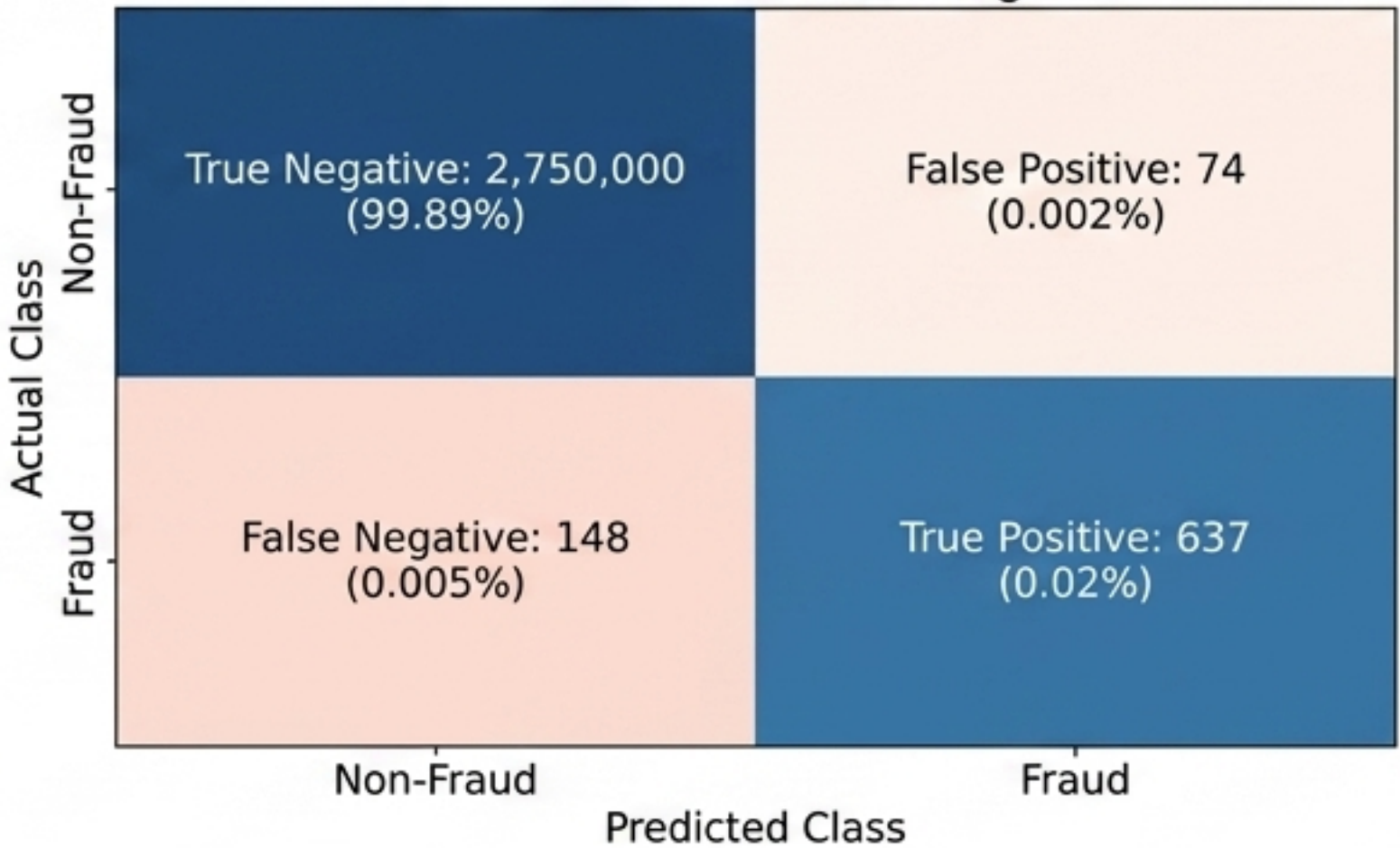
Precision: 97.37%

When the system flags fraud, it is almost always right

Recall: 81.14%

Successfully isolates the vast majority of financial threats

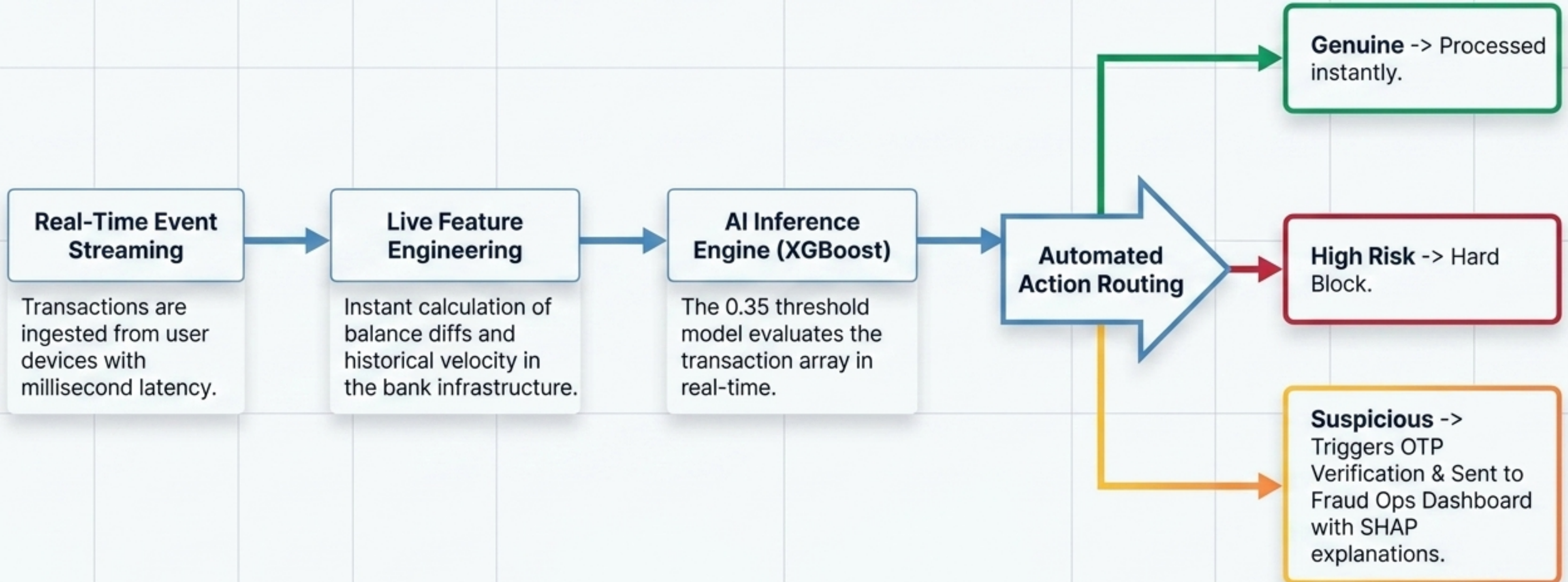
Confusion Matrix Heat-grid



Business Translation

The system achieves an elite tier of **accuracy (99.9%)**, practically eliminating false alarms while providing a massive upgrade in **threat detection** over legacy rules.

Proposed Production Deployment Architecture



Business Recommendations & Conclusion

Strategic Consulting Recommendations

Targeted Real-Time Monitoring

Deploy hyper-vigilance protocols specifically on CASH_OUT and TRANSFER endpoints.

Frictionless Interventions

Implement dynamic OTP step-ups exclusively for transactions breaching the 0.35 risk threshold.

Continuous MLOps

Establish automated model retraining pipelines to adapt to evolving adversary tactics.

Value Delivered Summary

- ✓ **Technical Triumph:** Transitioned from rigid rules to a dynamic, explainable XGBoost risk engine.
- ✓ **Operational Efficiency:** SHAP integration drastically cuts investigation time and false-positive fatigue.
- **Bottom Line:** A deployment-ready architecture that actively reduces financial loss while protecting the customer experience.